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5. Himalayan balsam *(Impatiens glandulifera)* is a non-native, invasive species found in the UK. Himalayan balsam grows rapidly, out-competing many local species.

Himalayan balsam in flower

In laboratory trials, scientists found that a fungus could kill Himalayan balsam.

Leaf of Himalayan balsam infected with fungus

Scientists investigated the possibility of using fungus to reduce the growth of the invasive plant on a larger scale.

(a) State the term that describes the use of a living organism to reduce the growth of an invasive species. [1]

....

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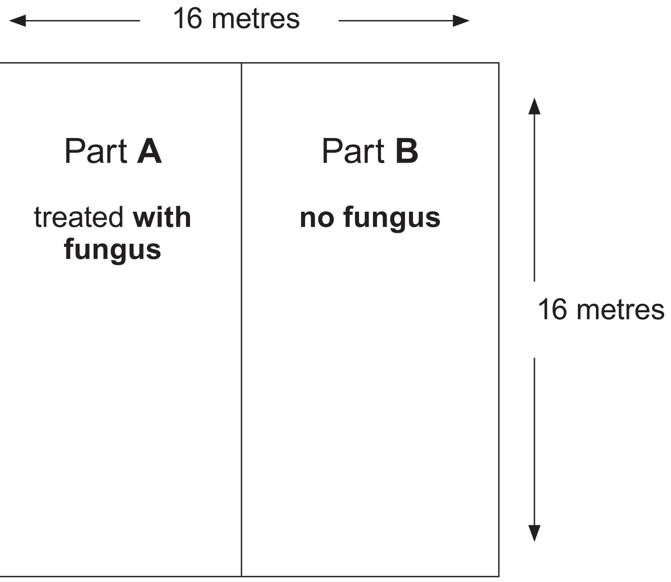
(b) The scientists marked out an area, 16 metres × 16 metres, on which many Himalayan balsam plants were growing and were evenly distributed.

They then applied the fungus to half of the section (part **A**). They did not apply any treatment to the other half (part **B**).

Plan of field trial



After six weeks, the scientists estimated the total number of Himalayan balsam plants by quadrat sampling in parts **A** and **B**.

(i) State why it was important that the quadrats were placed at random. [1]

....

....

(ii) Describe a sampling technique they would carry out, using 1 m^2 quadrats. [3]

....

....

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....

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(c) For part **A**, the total number of Himalayan balsam plants found in 12 quadrats was 36.

Calculate

(i) the mean number of plants per 1 m^2 quadrat in part **A**. [1]

.... plants / m^2

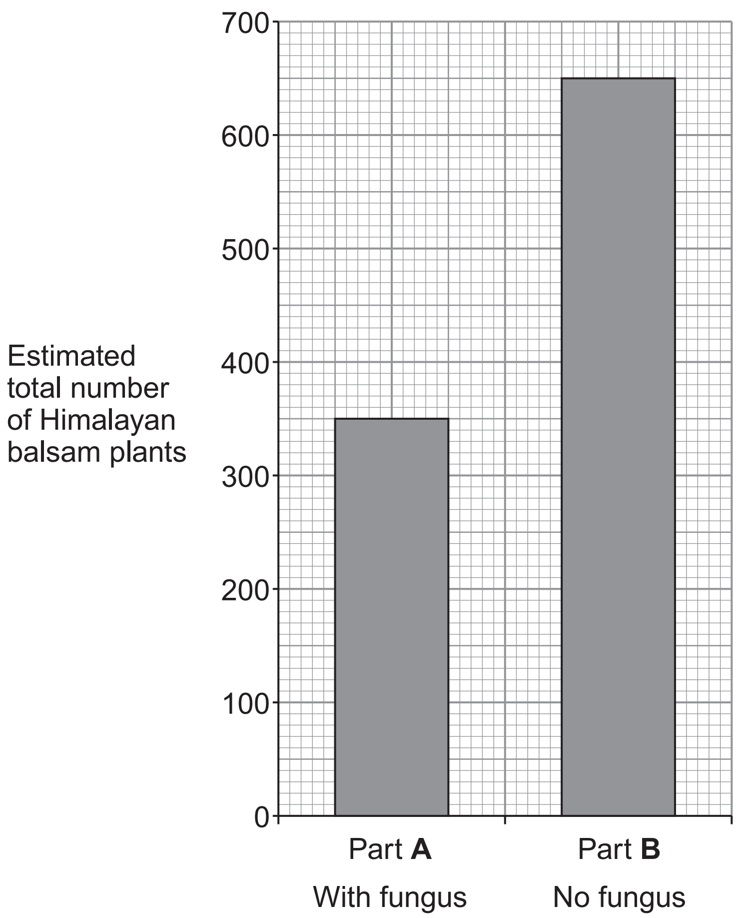
(ii) I. the area of part **A**. [1]

.... m^2

II. the estimated total number of plants in part **A.** [1]

.... plants

(d) The scientists carried out their investigation a further 5 times to be confident that their results were repeatable. The mean results of all their investigations are shown in the bar chart.



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(i) State the conclusion that could be drawn from these results. [1]

....

....

(ii) State how the scientists could test that the results were reproducible. [1]

....

....

(e) (i) The scientists went on to investigate the effects of the fungus on many native plant species. State why this was necessary. [1]

....

....

(ii) State how biodiversity would be affected if the Himalayan balsam plants were not controlled. [1]

....

....

12